

Digital Twin Development for Viper 650 and Scara Substations in Smart Manufacturing Assembly Line

Sujal S Patil

Automation and Robotics Engineering
KLE Technological University, Hubballi,
India.

sujalpatil203@gmail.com

Sneha Uppar

Automation and Robotics Engineering
KLE Technological University, Hubballi,
India.

snehauppar45@gmail.com

Abidali Mujawar

Automation and Robotics Engineering
KLE Technological University, Hubballi
India.

aliabidshama@gmail.com

Sudeep Talwar

Automation and Robotics Engineering
KLE Technological University, Hubballi,
India

talwarsudeep2@gmail.com

Vinit V Bangare

Automation and Robotics Engineering
KLE Technological University, Hubballi,
India.

vinitbangare632@gmail.com

Mr. Girish Karikatti

Automation and Robotics Engineering
KLE Technological University, Hubballi
India.

girish_k@kletech.ac.in

Dr.Madhusudhan H.k.

Automation and Robotics Engineering
KLE Technological University, Hubballi,
India.

madhusudhana@kletech.ac.in

Abstract—Digital twin technology has emerged as a key enabler in modern smart manufacturing, providing a virtual representation of physical assets to support real-time monitoring, virtual commissioning, and process optimisation. This work focuses on the development of a digital twin for an automated manufacturing system consisting of an Omron Viper 650 industrial robot, a SCARA robot, and an Automated Storage and Retrieval System (ASRS). The proposed system integrates multiple platforms, including Sysmac Studio for PLC logic development, Schneider ControlBuild for communication modelling and Functional Mock-up Unit (FMU) export, and the DELMIA 3DEXPERIENCE platform for 3D simulation and virtual validation.

Through this integration, a bidirectional communication framework was established, enabling the PLC to control the virtual system while receiving real-time feedback from the digital twin. This allowed accurate synchronisation between the virtual and physical environments. Experimental validation demonstrated that the behaviour of the digital twin closely matched the real system, with minimal deviations in timing and sequence execution. The use of the digital twin significantly improved the commissioning process by identifying control logic errors, motion planning issues, and potential safety risks prior to physical deployment.

Furthermore, the integration of the SCARA robot enhanced highspeed pick-and-place operations, while the ASRS improved material handling and storage efficiency within the system. Overall, the proposed framework demonstrates a reliable and scalable approach for integrating industrial robots, SCARA systems, and automated storage solutions using digital twin technology, contributing to improved efficiency and reduced commissioning time in smart manufacturing environments.

Index Terms—Digital twin, Virtual commissioning, Smart manufacturing, SCARA robot, ASRS, Sysmac Studio, ControlBuild, DELMIA, OPC UA, FMU.

I. INTRODUCTION

Manufacturing systems are undergoing a major transition as industries move toward highly connected, automated, and intelligent production environments. This transformation, often linked to the principles of Industry 4.0, places a strong focus on data-driven decision-making, real-time control, and seamless interaction between digital and physical systems. Among the technologies enabling this shift, the concept of the digital twin has gained significant importance. A digital twin goes beyond a basic simulation—it is a digital counterpart of a physical asset that continuously updates through live data exchange, allowing the virtual model to reflect the real system's condition and behaviour almost instantly. With this capability, engineers can analyse, test, and optimise operations without interfering with the real equipment.

In conventional automation projects, commissioning is one of the most challenging and time-consuming phases because hardware, control logic, sensors, and robotic systems must function together for the first time under real conditions. Errors such as mismatched timing, inaccessible robot positions, and unexpected equipment interactions often surface only after installation, leading to downtime, increased labour, and potential mechanical stress. Virtual commissioning enabled through digital twin technology provides an effective solution by enabling full validation of control logic and robotic motion in a risk-free environment before operating the physical system. The present work

focuses on developing a digital twin for an automated manufacturing cell consisting of multiple integrated systems, including the Viper 650 industrial robot, a SCARA robot, and an Automated Storage and Retrieval System (ASRS). The Viper 650 robot is designed for high-speed and high-precision industrial applications, while the SCARA (Selective Compliance Assembly Robot Arm) robot is widely used for pick-and-place and assembly tasks due to its high speed, repeatability, and efficient planar motion. The ASRS further enhances the system by automating material storage and retrieval, improving overall workflow efficiency and reducing manual handling. Although these systems serve different functions, they require synchronised control, efficient communication, and reliable motion planning to operate as a unified system. The objective of this project is to develop a virtual representation that accurately replicates the real system and enables complete operational testing without physical execution.

To achieve this, a multi-platform workflow was implemented using Omron Sysmac Studio, Schneider ControlBuild, and the DELMIA 3DEXPERIENCE platform. In this configuration, Sysmac Studio was responsible for PLC-based control logic, ControlBuild acted as middleware for communication modelling and Functional Mock-up Unit (FMU) export, and DELMIA provided the 3D simulation environment for validating robot movement and overall system behaviour. This integration established a two-way connection between the real and virtual systems, allowing PLC logic to control the digital twin while the virtual environment returned real-time feedback signals.

By enabling complete verification prior to deployment, the digital twin helped identify sequence-related errors, path planning issues, and potential safety risks without exposing the physical system to damage or downtime. The approach proved highly effective for debugging and optimisation and demonstrated that industrial robots, SCARA systems, and automated storage solutions can significantly benefit from digital twin-based virtual commissioning in modern smart manufacturing environments.

II. LITERATURE REVIEW

A. Digital Twin Architecture for Smart Manufacturing Systems

Digital Twin technology has surfaced as a fundamental facilitator of cyber-physical production systems in Industry 4.0. Initial research by Uhlemann et al. established basic structures linking tangible manufacturing resources with digital representations via ongoing data communication [1]. Qi and Tao further proposed structured Digital Twin frameworks that integrate physical entities, virtual models, services, data, and communication links [2].

These architectures usually consist of physical components, virtual representations, data handling layers, services, and communication connections to facilitate real-time oversight and management. Later studies expanded these architectures by incorporating simulation, analytics, and decision-support components for manufacturing optimisation [4], [6]. These frameworks enable synchronisation among machines, sensors, and digital models to enhance production visibility and forecasting abilities.

Several review studies also emphasise the importance of IoT platforms, cloud computing, and big data technologies in facilitating scalable Digital Twin environments [5], [27], [30]. Nevertheless, many initial frameworks remained theoretical and did not provide comprehensive implementation plans for industrial automation settings. Minimal focus was placed on PLC-based systems and the communication protocols necessary for real-time synchronisation. Consequently, effective implementation in manufacturing assembly lines necessitates additional architectural improvements.

B. Real-Time Data Integration and Communication in Digital Twin Systems

Immediate interaction between tangible manufacturing systems and digital representations is vital for successful Digital Twin execution. Studies conducted by Tao and Qi highlight the importance of continuous data synchronisation between physical devices and digital models to support intelligent manufacturing operations [3].

Additional research also emphasises the role of industrial networks and IoT technologies in enabling reliable data exchange within manufacturing systems [30]. OPC UA is recognised as a valuable communication standard because of its reliability and interoperability in industrial automation environments. Chen et al. demonstrated the application of OPC UA communication for real-time monitoring of manufacturing processes through Digital Twin models [22].

Cloud-based systems and edge computing have also been suggested to facilitate decentralised data processing and realtime analytics for industrial applications [11], [17]. Even with these progressions, numerous studies offer only broad communication models lacking specific execution for PLC-driven assembly lines. Challenges like communication delay, data compatibility, and network dependability continue to be active areas of research. Consequently, strong industrial communication frameworks are still required for extensive Digital Twin implementation.

C. Digital Twin Applications in Smart Assembly and Robotic Manufacturing

Digital Twin technology has been extensively examined to enhance efficiency in robotic and automated assembly

systems. Numerous research works concentrate on merging digital models with robotic workstations to assess production efficiency and enhance manufacturing processes.

For example, Zhuang et al. suggested a production management framework based on Digital Twin technology for intricate assembly systems [10]. Recent studies also demonstrate how synchronised digital models can support dynamic scheduling and predictive analysis in robotic assembly lines [9], [12]. These systems allow engineers to model production situations and modify operational settings instantly.

Moreover, Digital Twin models facilitate adaptable manufacturing by allowing swift responses to process disruptions or equipment malfunctions. Cloud-based Digital Twin platforms have also been explored for the remote monitoring and control of robotic manufacturing systems [17]. Nevertheless, numerous implementations still focus solely on simulation or particular case studies. Collaboration with industrial controllers and large-scale assembly systems has not been thoroughly confirmed. Consequently, additional investigation is needed to create scalable Digital Twin models for intelligent assembly lines.

D. Digital Twin-Based Virtual Commissioning and System Validation

Virtual commissioning has emerged as a crucial application of Digital Twin technology in the development of manufacturing systems. Engineers can assess control logic and system performance by generating digital replicas of machines and production processes prior to physical implementation.

Several studies demonstrate how simulation models can be integrated with automation systems to validate manufacturing processes and reduce commissioning risks [7], [13], [29]. This method greatly shortens commissioning duration, minimises implementation risks, and decreases operational mistakes. Virtual environments based on Digital Twins enable engineers to evaluate various production scenarios and enhance system setups.

Other studies further highlight the advantages of behavioural Digital Twin models for validating control strategies during the design stage [15], [20]. Even with these benefits, the majority of studies emphasise simulation environments instead of real-time engagement with actual production systems. The connection with PLC controllers and industrial communication standards is still restricted. Additionally, scalability for extensive manufacturing facilities has not been thoroughly explored. As a result, additional studies are necessary to create completely integrated virtual commissioning frameworks.

E. Challenges and Research Gaps in Industrial Digital Twin Implementation

Despite the considerable promise of Digital Twin technology for smart manufacturing, numerous technical hurdles persist. Many current studies mainly concentrate on conceptual frameworks or validation through simulation instead of real industrial implementation [26], [27], [30].

Instantaneous synchronisation between physical systems and digital models necessitates dependable communication frameworks and efficient data processing abilities. Another significant challenge is the integration of legacy manufacturing systems and PLC-based automation platforms, especially in brownfield production settings [19]. The performance of Digital Twin implementations is also influenced by data interoperability, communication latency, and system scalability.

Moreover, only a small number of studies offer experimental validation in complete assembly lines [5], [30]. Sophisticated applications like predictive maintenance, production scheduling, and adaptive control necessitate strong real-time data integration systems. Innovative technologies like edge computing, cloud systems, and fast industrial networks could assist in overcoming these constraints [11], [17].

Upcoming studies need to emphasise practical application frameworks that integrate Digital Twins with PLC control systems and industrial communication protocols like OPC UA [22]. Such advancements will be crucial for attaining completely synchronised smart manufacturing systems.

III. METHODOLOGY

A. Hardware Architecture and Design

The smart manufacturing workstation developed in this project is built around multiple integrated subsystems, including the Omron Viper 650 industrial robot, a SCARA robot, and an Automated Storage and Retrieval System (ASRS). These systems work together to perform coordinated material handling, storage, and transfer operations within the manufacturing cell. All components are controlled using an Omron NX-series PLC programmed in Sysmac Studio, which serves as the central controller for task sequencing, motion execution, gripper control, and safety management.

The PLC processes signals from various sensors distributed across the workstation, including proximity sensors for part detection, gripper-status sensors for confirming successful handling, and position sensors for ASRS operation. These inputs enable real-time decision-making and adaptive system control based on actual operating conditions.

Both robotic systems are equipped with pneumatic grippers to handle light-to-medium-weight components. The grippers are actuated through solenoid valves connected to PLC digital outputs, while feedback signals are transmitted to the PLC through digital inputs. To ensure consistency between the

real and virtual environments, all physical parameters—such as robot base positions, working heights, approach paths, and pick-and-place coordinates—were accurately replicated in the digital twin model.

The Viper 650 robot is utilised for high-speed and highprecision pick-and-place operations, while the SCARA robot is responsible for fast assembly and horizontal transfer tasks due to its high repeatability and planar motion efficiency. The ASRS enhances the system by automating part storage and retrieval, ensuring smooth material flow and reducing manual intervention. Together, these subsystems form a coordinated automation environment focused on efficiency, synchronisation, and reliability.

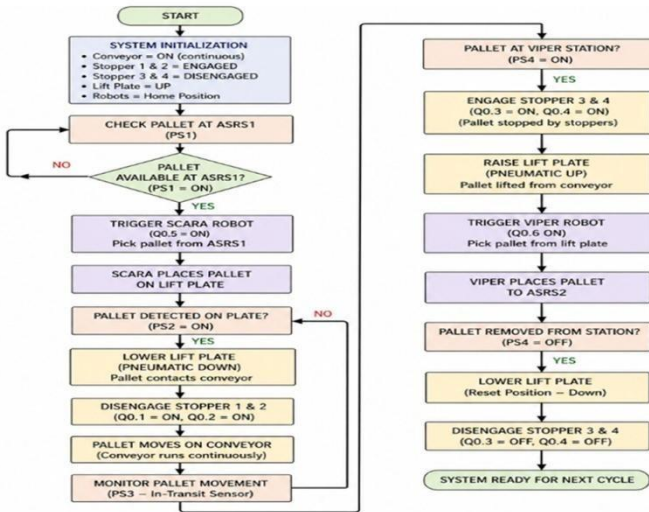


Fig. 1 Program Flowchart

B. Phase 1: Digital Cell Development in DELMIA

The first phase involved developing a realistic digital model of the manufacturing cell in the DELMIA 3DEXPERIENCE platform. This included:

- Importing and configuring 3D models of the Viper 650, SCARA robot, and ASRS
- Defining robot joint limits, work envelopes, and reachability zones
- Teaching approach, pick, and placement points for robotic operations
- Simulating gripper actuation (open/close actions)
- Creating collision zones and safe motion trajectories

Each subsystem was initially simulated independently to verify reachability, motion feasibility, and collision-free operation. At this stage, DELMIA functioned purely as a simulation tool without interaction with the PLC.

This phase ensured that robot paths, tool centre points (TCP), payload handling, and cycle-time estimates were validated before integrating the control logic.

C. Phase 2: Hardware–Software Bridging using ControlBuild

The next phase focused on establishing communication between the PLC and the virtual environment. Schneider ControlBuild was used as an interface layer to manage signal exchange and prepare the system for real-time interaction. Key activities included:

- Importing Sysmac PLC network variables using OPC UA
- Classifying variables into command, status, and safety categories
- Ensuring consistent naming conventions between Sysmac and ControlBuild
- Simulating I/O behaviour to validate logical flow
- Resolving data-type mismatches and update-rate inconsistencies

After successful validation, a Functional Mock-up Unit (FMU) was generated. This FMU encapsulated the control logic and signal flow required for integration with DELMIA, effectively acting as a virtual PLC interface for the digital twin.

D. Phase 3: FMU Integration and Real-Time Synchronisation

The FMU was imported into the DELMIA environment to enable real-time interaction between the PLC logic and the digital twin:

- Mapping DELMIA I/O points (e.g., Robot Start, Gripper Close, Part Detected, ASRS Load) to FMU signals
- Establishing bidirectional communication between the PLC and DELMIA
- Enabling real-time feedback from virtual sensors and actuators

Once integrated, DELMIA transitioned from a static simulation to a fully functional digital twin. Robot operations and ASRS actions were driven directly by PLC logic, while virtual sensor feedback influenced PLC decision-making.

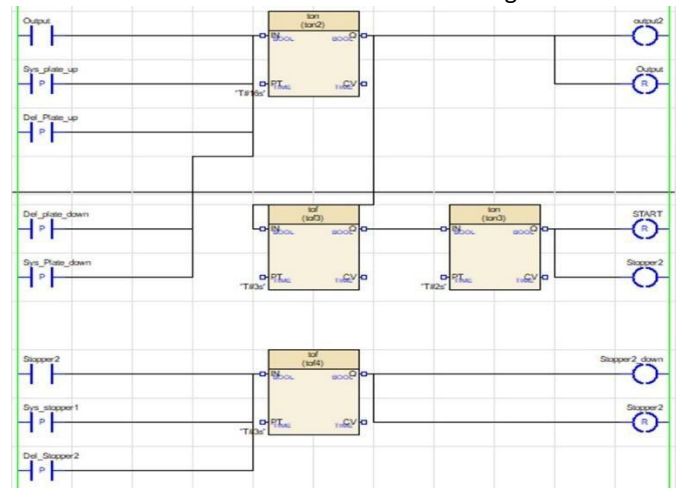


Fig. 2 Control Build Logic

The same PLC program used in the digital environment was later deployed to the physical system without modification, demonstrating high fidelity between virtual and real systems.

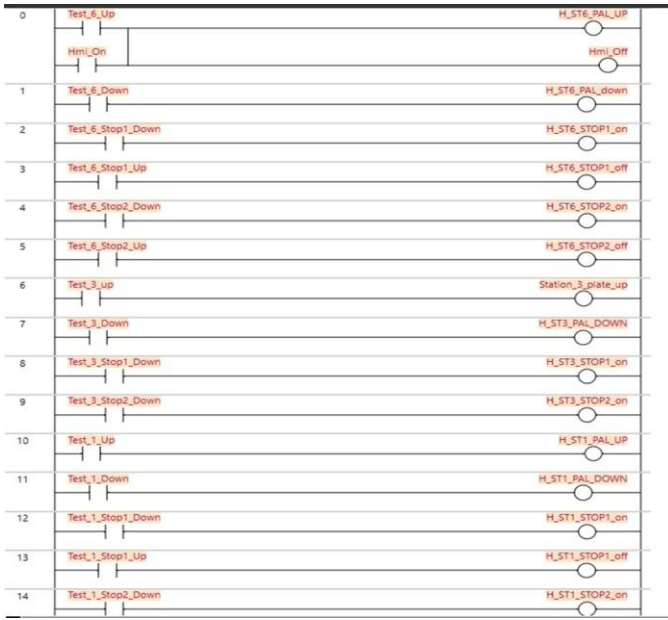


Fig. 3 Sysmac Studio Logic

E. Debugging and Challenges

During integration, several technical challenges were encountered:

- Signal mismatches between Sysmac and ControlBuild, resolved through strict naming conventions
- Communication delays due to update-rate conflicts, corrected by synchronising scan cycles
- FMU mapping errors caused by improper I/O grouping, resolved through structured signal organisation
- Collision detections in DELMIA, leading to optimised robot trajectories
- Synchronisation issues between robots and ASRS, resolved using event-based control logic

Addressing these issues in the virtual environment significantly reduced the risk of hardware damage and minimised physical debugging time.

F. Validation and Testing

System validation was performed in two stages:

1. Virtual Validation (FMU-Based Mode) The digital twin operated entirely under PLC control. All sequences—including robot motion, gripper actions, and ASRS operations—were successfully executed with proper synchronisation.

2. Physical Deployment

The same PLC program was deployed to the real system without modification. The physical system exhibited behaviour closely matching the digital twin, with timing deviations of approximately 1–2%.

The system operated smoothly without collisions or safety issues, confirming the accuracy of the digital twin.

G. Summary

The methodology demonstrates a structured approach to developing a digital twin for a multi-system manufacturing cell using DELMIA, ControlBuild, and Sysmac Studio. By validating control logic, robotic motion, and system synchronisation in a virtual environment, the approach significantly reduced commissioning time and improved system reliability. The framework is scalable and can be extended to include additional automation components such as conveyors, inspection systems, or multiple robotic stations.

IV. PROCESS CYCLE AND SYNCHRONISATION STRATEGY

The process cycle was designed to ensure reliable coordination between the Viper 650 robot, SCARA robot, and ASRS system. A signal-based synchronisation strategy was implemented, where each operation is triggered only after receiving confirmation from the previous step.

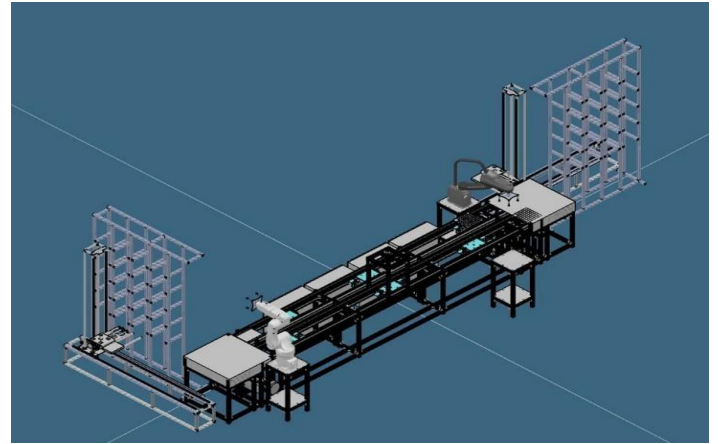


Fig.4 Virtual Prototyping(Delmia)

A. Sequence of Operation

- 1) Part Detection: A proximity sensor detects the arrival of a part and sends a signal (Part Detected) to the PLC.
- 2) Viper 650 Pickup: The Viper robot picks the part and sends a confirmation signal (Viper Grip OK).
- 3) Transfer to SCARA / ASRS: The part is transferred either to the SCARA robot for further processing or directly to the ASRS for storage, depending on system logic.
- 4) SCARA Operation: The SCARA robot performs pick-and-place or assembly operations and confirms completion (SCARA Done).
- 5) ASRS Storage/Retrieval: The ASRS stores or retrieves parts as required and signals completion (ASRS Done).

B. I/O Signal Mapping

TABLE I
I/O SIGNAL MAPPING

Signal Type	PLC Tag	Sender	Receiver	Purpose
Command	Viper Start	PLC	Viper	Start pickup
Status	Viper Grip OK	Viper	PLC	Confirm grip
Command	SCARA Start	PLC	SCARA	Start operation
Status	SCARA Done	SCARA	PLC	Operation complete
Command	ASRS Load	PLC	ASRS	Store part
Status	ASRS Done	ASRS	PLC	Storage complete
Safety	Cell Alarm	Any	PLC	Emergency stop

V. RESULTS

The digital twin implementation delivered highly positive results for the integrated manufacturing system consisting of the Viper 650 industrial robot, SCARA robot, and ASRS subsystem. After completing the multi-platform integration, the DELMIA virtual models responded directly to the Sysmac PLC logic in real time through the FMU generated in ControlBuild. This ensured that all subsystems operated without the need for simulation scripts or manual intervention, confirming that the digital twin functioned as a true extension of the physical automation system.

The complete process workflow—including part detection, robotic gripping, transfer between stations, SCARA-based handling operations, and ASRS storage/retrieval—was executed accurately in the virtual environment using PLC signals alone. During virtual commissioning, several issues that typically arise during physical testing—such as early gripper activation, minor path interference between robots, synchronisation delays, and inefficient ASRS response timing—were identified and resolved within the digital twin before deploying the system physically.

Due to these early corrections, the PLC program developed in Sysmac Studio was transferred directly to the real hardware without any modifications. During physical execution, the Viper 650, SCARA robot, and ASRS system followed the same sequence and operational logic validated in DELMIA. The system operated smoothly, with consistent synchronisation, no safety violations, and no operational conflicts between subsystems. The physical system's timing behaviour closely matched the digital twin, with only minor variations observed.

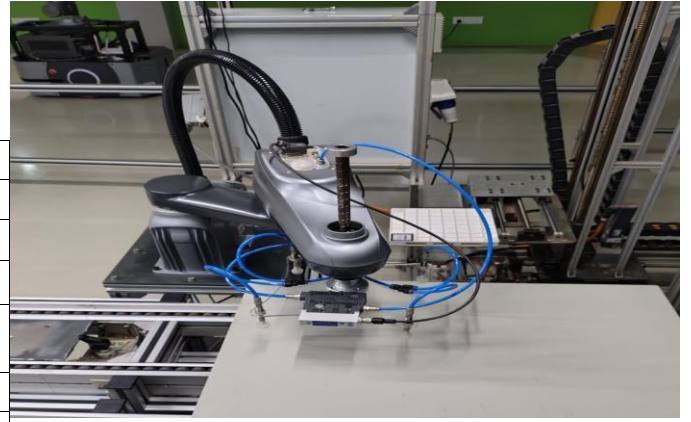


Fig.5 SCARA Robot



Fig. 6 Viper 650

Fig. 1. Complete digital twin workflow: (a) virtual commissioning in DELMIA 3DEXPERIENCE, (b) PLC motion logic development in Sysmac Studio, (c) I/O mapping and FMU integration in ControlBuild, and (d) physical execution of the robotic workstation with SCARA and ASRS integration

Fig. 5.1 illustrates the complete digital twin workflow, including: (a) virtual commissioning in DELMIA 3DEXPERIENCE, (b) PLC motion logic development in Sysmac Studio, (c) I/O mapping and FMU integration in ControlBuild, and (d) physical execution of the robotic workstation with SCARA and ASRS integration.

Although the figure primarily highlights one subsystem view, the same workflow applies to all integrated components, including the Viper 650, SCARA robot, and ASRS. This confirms that the digital twin framework accurately represents the behaviour of the entire system and ensures reliable operation across all subsystems.

VI. DISCUSSION

The results of the digital twin implementation clearly demonstrate the effectiveness of virtual commissioning in smart manufacturing systems involving multiple automated subsystems. One of the key observations from this work is that

several critical issues—such as motion timing mismatches, premature gripper activation, workspace interference, and



Fig. 7 Physical Smart Assembly Line.

ASRS coordination delays were successfully identified and resolved within the virtual environment before physical deployment. This significantly reduced the need for trial-and-error testing on actual hardware, minimising both risk and commissioning time.

The success of the system can be attributed to the seamless integration between Sysmac Studio, ControlBuild, and DELMIA. Robot motion and ASRS operations in DELMIA were driven entirely by PLC logic rather than pre-defined simulation scripts. This ensured that all improvements made during virtual commissioning were directly applicable to the real system. As a result, the traditional iterative process of modifying PLC programs after physical deployment was eliminated, leading to a more efficient development cycle.

The integration of the SCARA robot and ASRS subsystem further demonstrated the scalability of the digital twin approach. The SCARA robot provided high-speed and precise handling for intermediate operations, while the ASRS system enabled automated storage and retrieval, improving material flow and system efficiency. The ability to validate the interaction between these subsystems within the digital environment ensured smooth coordination and prevented operational conflicts in the real system.

An important insight from this work is that the accuracy of a digital twin depends heavily on consistent data handling and system modelling across platforms. Issues such as variable mismatches, data-type inconsistencies, and communication delays initially affected synchronisation between systems. Additionally, accurate modelling of robot kinematics, tool centre points (TCP), and ASRS movement parameters in DELMIA was essential to ensure that the virtual system closely matched real-world behaviour. These findings highlight that successful digital twin implementation requires both precise simulation modelling and robust communication architecture.

Overall, the results confirm that digital twin-based virtual commissioning significantly reduces commissioning effort,

minimises the risk of equipment damage, and accelerates system deployment. Beyond validation, the digital twin also served as an effective analytical tool for evaluating system performance, optimising motion strategies, and verifying safety mechanisms. This demonstrates that digital twins provide value throughout the entire lifecycle of smart manufacturing systems, making them an essential component of modern industrial automation. Compared to traditional commissioning, the proposed approach significantly reduced debugging time and eliminated repeated hardware trials.

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